

Riemannian versus KL Discretization of the Gaussian Natural Gradient Flow under the Improved KL Stepsize Bound

Abstract

The Gaussian natural gradient flow for variational inference, $\dot{m} = Cg$, $\dot{C} = C + CHC$, can be time-discretized in (at least) two ways that share the same explicit mean update but differ in the covariance update: a *Riemannian-distance* (geodesic / exponential-map) step and a *KL/Bregman* (rational inverse-covariance) step. An earlier analysis of the KL scheme carried a sufficient stepsize condition markedly smaller than that of the Riemannian scheme, through an obsolete cubic covariance-conditioning factor; the improved KL proof *removes* this penalty. Both schemes now share the same theorem-safe stepsize scale $\Delta t \leq 1/(\beta\lambda_{\max})$, and the only remaining theoretical difference between them is the per-step contraction factor, not the admissible stepsize. The numerical study therefore changes its question: how sharp is the improved (now shared) bound, how does the observed contraction compare against the theorem factors q_{Riem} and q_{KL} , and how do the two schemes compare at matched stepsizes? On three deterministic two-dimensional targets—an exact Gaussian posterior, a non-separable quartic log-concave posterior, and a smooth strongly log-concave posterior—and a scalar diagnostic, we classify every (target, λ , method, Δt) run as SPD-feasible, stable, monotone, or accurate. On the smooth targets the two schemes share the theorem-safe stepsize exactly, and both remain stable and monotone well beyond it, with comparable empirical/theory ratios—there is no KL-specific stepsize disadvantage from the theory. A scalar diagnostic isolates the failure mechanism: the KL covariance update is unconditionally well posed for log-concave targets, and the large-stepsize failures of *both* schemes are driven by the shared explicit mean update, not by the covariance step. A convergence-speed study across all three targets, three conditionings λ , and every convergent stepsize (Sec. 6) shows that at fine stepsizes the two schemes converge at essentially the same rate in elapsed time $n\Delta t$, while at the coarsest convergent stepsizes the Riemannian update converges faster per step on the smooth targets and KL is faster on the non-smooth quartic. Finally, on the common theorem-safe reference stepsize $\Delta t_{\text{ref}} = 1/(\beta\lambda_{\max})$ we benchmark the theorem contraction factors of both schemes against the observed contraction (Sec. 7): both are now genuine in-theorem rates, the observed contraction is faster than either, and q_{KL} is the more conservative of the two—a contraction-rate gap, not a stepsize gap.

1 Introduction

We study the Gaussian variational inference problem $\min_{m,C} \mathcal{E}(a)$ with $\mathcal{E}(a) = \text{KL}(\rho_a \parallel \rho_{\text{post}})$, $\rho_a = \mathcal{N}(m, C)$, $a = (m, C)$, $C \in \text{SPD}(d)$, and the natural gradient flow that minimizes it in the Fisher–Rao geometry. Writing

$$g_n = \mathbb{E}_{\rho_{a_n}}[\nabla_{\theta} \log \rho_{\text{post}}(\theta)], \quad H_n = \mathbb{E}_{\rho_{a_n}}[\nabla_{\theta}^2 \log \rho_{\text{post}}(\theta)], \quad (1)$$

the continuous flow is

$$\dot{m}_t = C_t g_t, \quad \dot{C}_t = C_t + C_t H_t C_t. \quad (2)$$

Two natural discretizations of (2) share the same explicit mean update and differ only in how the covariance is advanced. The Riemannian-distance discretization takes a true geodesic step

of the affine-invariant geometry; the KL/Bregman discretization takes a proximal/mirror step in the dually flat (Bregman) geometry of the exponential family. The two are equally faithful to the continuous flow to first order, but their finite-stepsizes behavior, and in particular the sufficient stepsize conditions under which the current analysis proves descent, differ substantially.

The earlier “tiny KL stepsize” question is now *obsolete*. A previous KL analysis admitted only $\Delta t_{\text{KL,theory}} = 1/(\beta\lambda_{\max})$ (an obsolete cubic factor), much smaller than the Riemannian $\Delta t_{\text{Riem,theory}} = 1/(\beta\lambda_{\max})$, and it was natural to ask whether the KL scheme truly needed such small stepsizes. The improved KL proof settles this on the theory side: it removes the cubic penalty, so the KL relative-smoothness constant is exactly $L_{\text{KL}} = \beta\lambda_{\max} = L_{\text{Riem}}$ and the two schemes share the theorem-safe stepsize

$$\Delta t_{\text{KL,theory}} = \Delta t_{\text{Riem,theory}} = \frac{1}{\beta\lambda_{\max}}, \quad \lambda_{\max} = \max\{\lambda_{0,\max}, 1/\alpha\}. \quad (3)$$

What remains genuinely different is the per-step *contraction factor* (Sec. 7), not the admissible stepsize. The study therefore now asks how sharp the improved (shared) bound is empirically, how the observed contraction compares against the theorem factors, and—at matched stepsizes—which scheme is more efficient on which target.

All experiments here are one- or two-dimensional, fully deterministic (closed-form or Gauss–Hermite expectations, no Monte Carlo), and CPU-only, so that the stepsize and stability conclusions are not confounded by sampling error.

2 Algorithms

Both schemes evaluate g_n and H_n once at the current state $a_n = (m_n, C_n)$ and share the explicit mean update

$$m_{n+1} = m_n + \Delta t C_n g_n. \quad (4)$$

They differ only in the covariance update.

Riemannian-distance discretization. A geodesic step of the affine-invariant geometry,

$$C_{n+1} = e^{\Delta t C_n^{1/2}} \exp(\Delta t C_n^{1/2} H_n C_n^{1/2}) C_n^{1/2}. \quad (5)$$

This is geometrically faithful—it is exactly the exponential map of the metric along the natural-gradient direction—and it preserves positive definiteness automatically, since it conjugates a matrix exponential (always SPD) by $C_n^{1/2}$. Its cost is a symmetric eigendecomposition (for $C_n^{1/2}$ and the matrix exponential) per step; in the $d \leq 2$ regime studied here this is negligible, but in general it is the more expensive update.

KL/Bregman discretization. A proximal step in the dually flat geometry of the Gaussian exponential family,

$$C_{n+1} = (1 + \Delta t)(C_n^{-1} - \Delta t H_n)^{-1}. \quad (6)$$

This is computationally simpler—a single symmetric solve of the precision matrix $P_n = C_n^{-1} - \Delta t H_n$ —and exposes a structural fact we will use repeatedly: when the target is log-concave, $H_n = -\mathbb{E}[\nabla^2 V] \preceq 0$, so $P_n = C_n^{-1} - \Delta t H_n \succ 0$ for *every* $\Delta t > 0$ and the KL covariance update is unconditionally well posed and SPD. Any KL failure must therefore come from elsewhere in the iteration.

Implementation. Both covariance updates symmetrize the result, $C \leftarrow (C + C^\top)/2$, after every step. The Riemannian step uses the eigendecomposition of the symmetric argument for $C^{1/2}$ and $\exp$; the KL step forms C_n^{-1} by a Cholesky solve, builds P_n , and inverts it through its Cholesky factor. We record the extreme eigenvalues of C_n at every step and treat any NaN/Inf, loss of positive definiteness ($\lambda_{\min}(C_n) \leq 10^{-12}$), or energy explosion as a failure.

Consistency. Linearizing (5) and (6) in Δt gives, for both, $C_{n+1} = C_n + \Delta t (C_n + C_n H_n C_n) + O(\Delta t^2)$, matching $\dot{C}$ in (2); together with (4) both schemes are first-order consistent with the continuous flow. (The test suite verifies this numerically by finite differences.)

3 Targets

We use three two-dimensional targets and one scalar diagnostic. Throughout, $V = -\log \rho_{\text{post}}$ up to a constant, $g = -\mathbb{E}[\nabla V]$, $H = -\mathbb{E}[\nabla^2 V]$, and the energy minimized (the KL up to an a -independent constant) is

$$\mathcal{E}(a) = \mathcal{E}(m, C) = -\frac{1}{2} \log \det C + \mathbb{E}_{\mathcal{N}(m, C)}[V(\theta)], \quad (7)$$

whose gap $\mathcal{E}(a_n) - \mathcal{E}(a_\star)$ is the convergence diagnostic plotted throughout. The two-dimensional targets all use $\lambda \in \{0.01, 0.1, 1\}$ and the ill-conditioning is intentional: a wide range of covariance and curvature scales is exactly the regime in which the now-removed cubic KL penalty was largest, so it is where the equality of the two theorem-safe stepsizes is most informative.

Gaussian posterior (target A). $\Phi_R(\theta) = \frac{1}{2} \theta^\top Q \theta$ with $Q = \text{diag}(1, \lambda)$, so $\rho_{\text{post}} = \mathcal{N}(0, Q^{-1})$ and

$$g_n = -Q m_n, \quad H_n = -Q \quad (8)$$

are exact and state-independent. The optimum is exact, $m_\star = 0$, $C_\star = Q^{-1}$, and the energy gap is analytic,

$$\mathcal{E}(a_n) - \mathcal{E}(a_\star) = \frac{1}{2} [m_n^\top Q m_n + \text{Tr}(Q C_n) - \log \det(Q C_n) - 2]. \quad (9)$$

Initialization $m_0 = (10, 10)^\top$, $C_0 = \text{diag}(\frac{1}{2}, 2)$. With $\nabla^2 V \equiv Q$ the global constants are $\alpha = \min(1, \lambda)$, $\beta = \max(1, \lambda)$. This is the cleanest stepsize test: there is no quadrature or sampling error anywhere.

Non-smooth (quartic) log-concave posterior (target B). $\Phi_R(x, y) = \frac{(\sqrt{\lambda}x - y)^2}{20} + \frac{y^4}{20}$. This is convex but *not* globally smooth: $\nabla^2 V$ grows without bound through the quartic term, so no theorem-based global (α, β) exists and we set `theory_bound_available=false`; only empirical stepsize thresholds are reported. We refer to this target as *non-smooth log-concave* throughout, the defining property being the absence of a global smoothness constant. All required expectations close in the Gaussian moments of Y , $\mathbb{E}[Y^2] = m_y^2 + C_{22}$, $\mathbb{E}[Y^3] = m_y^3 + 3m_y C_{22}$, $\mathbb{E}[Y^4] = m_y^4 + 6m_y^2 C_{22} + 3C_{22}^2$, so g , H and $\mathbb{E}[V]$ are exact. Initialization $m_0 = (10, 10)^\top$, $C_0 = 4I$. The optimum $a_\star$ is found numerically (below).

Smooth strongly log-concave posterior (target C). $\Phi_R^{\text{smooth}}(x, y) = \frac{(\sqrt{\lambda}x - y)^2}{20} + \frac{\delta}{2}(x^2 + y^2) + \gamma \log \cosh(y)$ with $\delta = 0.05$, $\gamma = 1$. This shares the non-separable coupled term of target B but is globally strongly log-concave and smooth, with

$$\alpha = \delta, \quad \beta = \delta + \frac{1+\lambda}{10} + \gamma, \quad (10)$$

from the rank-one coupled quadratic (top eigenvalue $(1+\lambda)/10$), the δI ridge, and $0 \leq \gamma \operatorname{sech}^2(y) \leq \gamma$. Expectations of $\tanh(Y)$, $\operatorname{sech}^2(Y)$ and $\log \cosh(Y)$ under $Y \sim \mathcal{N}(m_y, C_{22})$ are computed by 80-node Gauss–Hermite quadrature (deterministic, no Monte Carlo). Initialization $m_0 = (10, 10)^\top$, $C_0 = 4I$.

Scalar Gaussian diagnostic (target D). $\rho_{\text{post}} = \mathcal{N}(0, 1)$, so $g = -m$, $H = -1$, and the covariance recursions reduce to the closed forms

$$\text{KL: } C_{n+1} = \frac{(1 + \Delta t)C_n}{1 + \Delta t C_n}, \quad \text{Riem: } C_{n+1} = C_n \exp(\Delta t(1 - C_n)), \quad (11)$$

with the shared mean step $m_{n+1} = (1 - \Delta t C_n)m_n$. The KL form gives $C_{n+1} - 1 = (C_n - 1)/(1 + \Delta t C_n)$, a strict contraction toward $C = 1$ for any $\Delta t > 0$. We sweep $C_0 \in \{0.01, 0.1, 10, 100\}$ and $\Delta t \in \{0.5, 1, 2, 5\}$, plus a mean-overshoot probe $m_0 = 1$, $C_0 = 100$.

Reference optimum. For the non-Gaussian targets B and C we minimize (7) over the Cholesky parameterization $C = LL^\top$, $L_{11} = e^{s_1}$, $L_{22} = e^{s_2}$, L_{21} free, with L-BFGS-B and analytic gradients (via the Bonnet/Price identities $\partial_m \mathbb{E}[V] = \mathbb{E}[\nabla V]$, $\partial_C \mathbb{E}[V] = \frac{1}{2} \mathbb{E}[\nabla^2 V]$) from several deterministic restarts. The fitted optima are stationary to $\|C_\star^{-1} + H_\star\|_F < 10^{-9}$; optimizer diagnostics are saved to `target_metadata.json`.

Continuous-time reference. For each target/ λ we also integrate (2) from the same initial condition with `solve_ivp` (Radau, $\text{rtol} = 10^{-10}$, $\text{atol} = 10^{-12}$), flattening the symmetric C state and symmetrizing the vector field. This continuous reference is used only for the discretization *accuracy* metric and the overlaid reference curves; the stability conclusions do not depend on it.

4 Stepsize criteria

We sweep $\Delta t \in \{0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10\}$ to a fixed terminal time $T = 15$, taking $N = \lceil T/\Delta t \rceil$ steps per run so that $N\Delta t = T$ (up to the final partial step): every run integrates to the same elapsed time T , and the elapsed time after step n is $n\Delta t$. The coarsest stepsizes ($\Delta t = 5, 10$) are deliberate very-coarse stability probes. Each run is classified from its *full* trajectory (the long-format CSV may be decimated for size, but classification always uses every step).

Definition 1 (Feasibility classes). With $\mathcal{E}_n = \mathcal{E}(a_n)$ and $\mathcal{E}_\star = \mathcal{E}(a_\star)$, a run is

- **SPD-feasible** if it has no NaN/Inf and $\lambda_{\min}(C_n) > 10^{-12}$ for all n ;
- **stable** if SPD-feasible, $\mathcal{E}_N < \mathcal{E}_0$, and there is no explosion, $\mathcal{E}_n - \mathcal{E}_\star \leq 10^3(\mathcal{E}_0 - \mathcal{E}_\star)$ for all n ;
- **monotone** if SPD-feasible and $\mathcal{E}_{n+1} \leq \mathcal{E}_n + 10^{-10}$ for all n ;
- **accurate** if SPD-feasible and the terminal discretization error against the continuous reference satisfies $\|m_N - m(T)\|_2 + \|C_N - C(T)\|_F / \|C(T)\|_F \leq 10^{-2}$.

For each (target, λ , method) we report the largest feasible stepsize in each class, $\Delta t_{\max}^{\text{SPD}}$, $\Delta t_{\max}^{\text{stable}}$, $\Delta t_{\max}^{\text{monotone}}$, $\Delta t_{\max}^{\text{accurate}}$ (a plain maximum over passing stepsizes, so a non-monotone feasibility boundary is captured faithfully).

Table 1: Per-target reference optimum $\mathcal{E}_\star$ and global strong-convexity / smoothness constants (α, β) . The non-smooth target is convex but not globally smooth, so no theorem-based (α, β) is reported.

target	λ	α	β	$\mathcal{E}_\star$	global smooth
Gaussian posterior	0.01	0.010	1.000	-1.3026	yes
Gaussian posterior	0.1	0.100	1.000	-0.1513	yes
Gaussian posterior	1	1.000	1.000	1.0000	yes
smooth log-concave	0.01	0.050	1.151	-0.5637	yes
smooth log-concave	0.1	0.050	1.160	-0.4933	yes
smooth log-concave	1	0.050	1.250	-0.0749	yes
non-smooth log-concave	0.01	–	–	-2.8316	no
non-smooth log-concave	0.1	–	–	-1.6803	no
non-smooth log-concave	1	–	–	-0.5290	no

Definition 2 (Theoretical stepsizes). For the globally smooth targets (A, C), set $\lambda_{\min} = \min\{\lambda_{0,\min}, 1/\beta\}$ and $\lambda_{\max} = \max\{\lambda_{0,\max}, 1/\alpha\}$ from the extreme eigenvalues of C_0 . Under the improved KL proof both schemes share one relative-smoothness constant $L = \beta\lambda_{\max}$, hence one theorem-safe stepsize

$$\Delta t_{\text{Riem, theory}} = \Delta t_{\text{KL, theory}} = \frac{1}{\beta\lambda_{\max}}. \quad (12)$$

The KL stepsize carries no dependence on $\lambda_{\min}$: the formerly used cubic factor has been removed. For the non-smooth target B (no global smoothness constant) no such bound is reported.

The diagnostic quantities are the ratios $\Delta t_{\max}^{\text{stable}}/\Delta t_{\text{theory}}$ and $\Delta t_{\max}^{\text{monotone}}/\Delta t_{\text{theory}}$: a ratio close to 1 means the (shared) proof bound is tight; a larger ratio means the bound is conservative. Because the two schemes now share Δt_{theory} , the comparison is direct: any difference in these ratios reflects how far each *scheme* can be pushed in practice, not a difference in the theorem-safe scale.

5 Results

Metadata. Table 1 lists the reference optimum $\mathcal{E}_\star$ and the global constants (α, β) for each target/ λ . The Gaussian and smooth targets are globally smooth (theory bounds available); the non-smooth target is not.

Convergence. Figures 1–2 show the energy gap $\mathcal{E}(a_n) - \mathcal{E}(a_\star)$ against elapsed time $n\Delta t$ for representative stepsizes $\Delta t \in \{0.1, 0.5, 1\}$. On the two globally smooth targets both schemes converge cleanly at these stepsizes; the curves of the two methods are visually close at matched Δt , since they agree to first order and the shared mean update dominates the transient. The Gaussian target reaches the analytic optimum to machine precision; the smooth target converges to the numerically optimized $a_\star$. On the non-smooth target, both schemes require much smaller stepsizes (Fig. 3), consistent with its unbounded curvature: the representative stepsizes shown there are the largest that converge for that target ($\Delta t \in \{0.005, 0.01, 0.02\}$), an order of magnitude below those of the smooth targets.

Stepsize stability. Figures 4–6 are the stability heatmaps: for each $(\lambda, \Delta t)$ a cell is green (stable and monotone), amber (stable but non-monotone), or red (unstable). For the two smooth targets the dotted line marks the (now common) theoretical stepsize $\Delta t_{\text{theory}} = 1/(\beta\lambda_{\max})$, which is identical

Table 2: Empirical vs theoretical maximum stepsize by target/ λ /method, against the shared theorem-safe stepsize $\Delta t_{\text{theory}} = 1/(\beta\lambda_{\text{max}})$, reporting the largest stable, monotone, and accurate stepsizes and the monotone/theory and accurate/theory ratios. The ratios are modest (10^0 – 10^2) and of the same order for KL and Riemannian; the gap survives the strict accuracy criterion, not just bare stability.

target	λ	method	Δt_{theory}	$\Delta t_{\text{max}}^{\text{stab}}$	$\Delta t_{\text{max}}^{\text{mono}}$	$\Delta t_{\text{max}}^{\text{acc}}$	$\frac{\text{mono}}{\text{theory}}$ / $\frac{\text{acc}}{\text{theory}}$
Gaussian	0.01	kl	0.01	2.000	2.000	1.000	200 / 100
Gaussian	0.01	riemannian	0.01	2.000	1.000	1.000	100 / 100
Gaussian	0.1	kl	0.1	2.000	2.000	1.000	20 / 10
Gaussian	0.1	riemannian	0.1	2.000	1.000	1.000	10 / 10
Gaussian	1	kl	0.5	1.000	1.000	1.000	2 / 2
Gaussian	1	riemannian	0.5	2.000	1.000	1.000	2 / 2
smooth	0.01	kl	0.0434	1.000	1.000	1.000	23 / 23
smooth	0.01	riemannian	0.0434	1.000	1.000	1.000	23 / 23
smooth	0.1	kl	0.0431	1.000	1.000	1.000	23.2 / 23.2
smooth	0.1	riemannian	0.0431	1.000	1.000	1.000	23.2 / 23.2
smooth	1	kl	0.04	1.000	1.000	1.000	25 / 25
smooth	1	riemannian	0.04	1.000	1.000	1.000	25 / 25
non-smooth	0.01	kl	–	0.020	0.020	0.020	–
non-smooth	0.01	riemannian	–	0.050	0.020	0.010	–
non-smooth	0.1	kl	–	0.020	0.020	0.020	–
non-smooth	0.1	riemannian	–	0.050	0.020	0.010	–
non-smooth	1	kl	–	0.020	0.020	0.020	–
non-smooth	1	riemannian	–	0.050	0.020	0.020	–

for the two methods. Two features are immediate. First, for both schemes the green/amber region extends well to the right of the dotted theoretical line—the empirical stability boundary sits near $\Delta t \approx 1$ – 2 while the shared sufficient bound is at $\Delta t \approx 10^{-2}$ – 10^{-1} . Second, the two methods’ boundaries sit at comparable stepsizes; the dotted line is the *same* for both. The same picture holds across all three λ and both smooth targets.

Theory vs empirical at the shared theorem-safe scale. Figure 7 and Table 2 compare, for the two smooth targets, the (shared) theoretical stepsize against the empirical stable, monotone, and accurate maxima. We emphasize the *monotone* and *accurate* classes over bare stability: monotonicity certifies genuine descent at every step, and accuracy certifies that the discrete trajectory still tracks the continuous flow to 10^{-2} , so both are stronger evidence than non-divergence alone. Because the two schemes now share Δt_{theory} , their empirical/theory ratios are directly comparable: both are modest factors of order 10^0 – 10^2 , of the same order for KL and Riemannian, and they persist under the strict accuracy criterion. There is no KL-specific blow-up of the ratio: the former four-to-six order separation was an artifact of the cubic penalty that the improved proof removes. The remaining gap—both schemes are stable a couple of orders of magnitude beyond the sufficient bound—measures how conservative the shared bound is, not any difference between the schemes.

Scalar diagnostic. Figure 8 traces the scalar covariance recursions (11) from $C_0 \in \{0.01, 0.1, 10, 100\}$ at $\Delta t \in \{0.5, 1, 2, 5\}$. The KL covariance update contracts to $C = 1$ for every C_0 and every stepsize shown—as the closed form $C_{n+1} - 1 = (C_n - 1)/(1 + \Delta t C_n)$ guarantees—never leaving the SPD cone. The Riemannian scalar update $C_{n+1} = C_n \exp(\Delta t(1 - C_n))$ can overshoot for large $C_0 \Delta t$.

The mean-overshoot probe ($m_0 = 1$, $C_0 = 100$) confirms the companion fact: the shared explicit mean step $m_{n+1} = (1 - \Delta t C_n)m_n$ amplifies the mean for both schemes once $\Delta t C_n > 2$ (e.g. $|m|$ grows past 10^4 at $\Delta t = 5$ even though the KL covariance remains perfectly stable). This locates the large-stepsize instability in the mean update shared by both schemes, not in the KL covariance step.

Cost. Since the dimension is $d \leq 2$, wall-clock differences between the schemes are small and noisy and we do not draw strong cost conclusions from them. Figure 9 reports iterations-to-tolerance (gap $\leq 10^{-4}$) for stable runs as the meaningful efficiency axis: at matched Δt the two schemes need a comparable iteration count, and the practical advantage of a scheme is governed by how large a stable Δt it admits rather than by per-step cost. The KL covariance step (one symmetric solve) is the cheaper update per step; the Riemannian step (eigendecomposition for $C^{1/2}$ and exp) is more expensive in higher dimension, though that difference is immaterial here.

6 Convergence speed: Riemannian versus KL at matched stepsize

The preceding sections concern *stability*: how large a stepsize each scheme tolerates and whether the KL proof bound is conservative. We now address the complementary and practically decisive question of *convergence speed*: given a stepsize that converges for both schemes, which scheme reaches the optimum faster, and how does that depend on the target, the conditioning λ , and the stepsize? This is the question a practitioner choosing between the two updates actually faces. Throughout we restrict to the convergent (stable) stepsizes for each (target, λ , method); divergent stepsizes are excluded by construction, since a diverging run has no meaningful convergence speed.

Summary: Riemannian usually wins at a matched stepsize. Table 3 summarizes the head-to-head comparison. For each target and stepsize it reports the terminal energy gap of each scheme (geometric mean over the three λ), the faster scheme, and the factor separating them. The pattern is clear and is the main practical message of this section. On the two *globally smooth* targets the Riemannian scheme wins at every stepsize, and its advantage *grows with the stepsize*: from a modest $1.3\text{--}7\times$ at the finest stepsizes to $10^4\text{--}10^8\times$ smaller terminal gap at the coarsest convergent stepsizes. Only on the non-smooth quartic target—whose unbounded curvature drives the Riemannian exponential map near its stability edge at the coarsest stable stepsize—does KL win, and there only by a modest factor. So in the regime where one routinely operates (a globally smooth target, a stepsize chosen as large as stability allows), the geodesic Riemannian update is the faster of the two, often by many orders of magnitude.

Energy gap at matched stepsizes. Figures 10–12 show the comparison panel by panel: one figure per target, with rows indexing the conditioning λ and columns indexing a representative convergent stepsize, so that each panel contains exactly two curves—Riemannian (solid) and KL (dashed)—at one $(\lambda, \Delta t)$ cell. The stepsizes shown are the curated convergent set for each target (stable for both schemes at every λ); the non-smooth target uses stepsizes an order of magnitude smaller, reflecting its unbounded curvature. On the two *globally smooth* targets (Figs. 10–11) the comparison splits cleanly by stepsize regime. At the *fine* stepsizes (left columns) the Riemannian and KL curves are nearly indistinguishable: the two schemes agree to first order, so when Δt is small they descend along essentially the same gap-versus- $n\Delta t$ curve, and that curve is the same across columns—the discrete trajectories track the common continuous flow. At the *coarse* stepsizes (right columns), however, the schemes separate visibly: the Riemannian update converges markedly faster per unit elapsed time $n\Delta t$, reaching machine precision while KL plateaus several orders of

Table 3: Head-to-head convergence speed at matched stepsize: terminal energy gap $\mathcal{E}(a_N) - \mathcal{E}(a_*)$ (geometric mean over $\lambda \in \{0.01, 0.1, 1\}$) for each scheme, the faster scheme, and the factor by which its terminal gap is smaller. Riemannian wins on both smooth targets, with the margin growing as Δt coarsens; KL wins only on the non-smooth quartic, and only modestly.

target	Δt	gapRiem	gapKL	winner	factor
Gaussian	0.05	2.5×10^{-11}	3.4×10^{-11}	Riem.	1.33
	0.1	1.3×10^{-11}	2.5×10^{-11}	Riem.	2
	0.5	3.4×10^{-15}	2.5×10^{-10}	Riem.	7.3×10^4
	1	1.3×10^{-16}	8.6×10^{-9}	Riem.	6.6×10^7
smooth	0.05	2.5×10^{-10}	6.8×10^{-10}	Riem.	2.67
	0.1	1.6×10^{-10}	1.2×10^{-9}	Riem.	7.12
	0.5	2.0×10^{-12}	4.2×10^{-8}	Riem.	2.2×10^4
	1	3.0×10^{-15}	1.2×10^{-6}	Riem.	4.1×10^8
non-smooth	0.002	2.1×10^{-9}	1.7×10^{-9}	KL	1.29
	0.005	1.2×10^{-9}	6.4×10^{-10}	KL	1.91
	0.01	9.2×10^{-11}	8.1×10^{-11}	KL	1.14
	0.02	9.4×10^{-6}	6.2×10^{-9}	KL	1.5×10^3

magnitude higher (at the Gaussian target, $\lambda = 0.01$, $\Delta t = 1$, the terminal gaps are $\sim 10^{-14}$ for Riemannian against $\sim 10^{-7}$ for KL). The geodesic covariance update therefore retains higher-order fidelity to the flow as the stepsize coarsens, whereas the rational KL update accumulates a larger discretization error—a genuine per-step convergence-rate advantage for the Riemannian scheme at large stepsizes, distinct from the stability question of the preceding sections. The non-smooth quartic target (Fig. 12) is the exception: there the two schemes agree at the finer stepsizes but *KL* is the faster scheme, decisively so at the coarsest convergent $\Delta t = 0.02$, where the Riemannian exponential map sits near its stability edge while the rational KL update has already converged. In all cases both schemes remain monotone and convergent throughout; the difference is the rate, not whether they converge.

The cost of a stepsize. Figure 13 quantifies the efficiency consequence. For the stable runs we measure the iterations and the elapsed time $n\Delta t$ required to reach a fixed energy-gap tolerance (10^{-6}), averaged over λ , as a function of Δt . The iteration count (top row) falls as $1/\Delta t$ on the log-log axes—a larger convergent stepsize reaches the same tolerance in proportionally fewer steps—while the elapsed time $n\Delta t$ to that tolerance (bottom row) is nearly flat in Δt until the largest convergent stepsizes, where the discrete trajectory begins to lag the continuous flow. Consistent with the matched-stepsize panels, the two schemes coincide at the fine and intermediate stepsizes and separate only at the coarsest convergent stepsizes, where the faster scheme reaches the tolerance in fewer iterations (and the slower one may not reach 10^{-6} at all within the horizon, terminating at a larger plateau). Which scheme that is depends on the target: on the two smooth targets it is Riemannian, while on the non-smooth target it is KL—at $\Delta t = 0.02$ the Riemannian run never reaches 10^{-6} whereas KL does. The practical reading is twofold: the dominant efficiency lever remains how large a stable stepsize a scheme admits (the conclusion of the stability sections), but among the stepsizes both schemes can take, the geometrically faithful Riemannian update converges faster per step in the coarse-stepsize regime on smooth targets, while the two are interchangeable when Δt is small.

7 Theorem contraction-rate benchmark at the shared theorem-safe scale

The stability sections established that the (now shared) theorem-safe stepsize is conservative, and the convergence-speed section compared the two schemes empirically at matched stepsizes. We now test the complementary question that carries the only remaining theoretical distinction: do the theorem-predicted *contraction factors* predict the observed contraction, and how much slack do they carry? Under the improved KL proof both factors are valid on the same theorem-safe range $\Delta t \leq 1/(\beta\lambda_{\max})$, so this is a genuine in-theorem comparison, not a formal extrapolation.

Why only the two globally smooth targets. We restrict to the exact Gaussian posterior (target A) and the smooth strongly log-concave posterior (target C), because both satisfy global α -strong log-concavity and β -smoothness, so a clean theorem contraction factor exists. We exclude the non-smooth quartic target B: it has no global β , so no theorem rate is available to benchmark against.

The shared theorem-safe reference stepsize. Both schemes share the theorem-safe reference stepsize

$$\Delta t_{\text{ref}} = \frac{1}{\beta\lambda_{\max}}, \quad \lambda_{\min} = \min\{\lambda_{0,\min}, 1/\beta\}, \quad \lambda_{\max} = \max\{\lambda_{0,\max}, 1/\alpha\}, \quad (13)$$

and we sweep the fractions $c \in \{0.05, 0.1, 0.25, 0.5, 0.75, 1\}$ with $\Delta t = c \Delta t_{\text{ref}}$, to the same terminal time $T = 15$ ($N = \lceil T/\Delta t \rceil$ steps). Under the improved KL proof the grid $\Delta t \leq \Delta t_{\text{ref}}$ lies inside the theorem range for *both* schemes, so $q_{\text{Riem}}(\Delta t)$ and $q_{\text{KL}}(\Delta t)$ are both genuine theorem contraction factors here—not a formal extrapolation onto an unproved range.

Contraction factors. For the Riemannian scheme the theorem per-step factor and per-unit-time rate are

$$q_{\text{Riem}}(\Delta t) = 1 - \alpha\lambda_{\min}\Delta t(2 - \beta\lambda_{\max}\Delta t), \quad r_{\text{Riem}}(\Delta t) = -\frac{\log q_{\text{Riem}}(\Delta t)}{\Delta t}. \quad (14)$$

On the grid $\Delta t \leq \Delta t_{\text{ref}}$ the bracket $(2 - \beta\lambda_{\max}\Delta t) \geq 1$, so $q_{\text{Riem}} \in (0, 1)$. The improved KL theorem gives the per-step factor and rate

$$q_{\text{KL}}(\Delta t) = 1 - \frac{\alpha\lambda_{\min}\Delta t}{2(1 + \Delta t)(1 + \Delta t\beta\lambda_{\max})}, \quad r_{\text{KL}}(\Delta t) = -\frac{\log q_{\text{KL}}(\Delta t)}{\Delta t}. \quad (15)$$

For reference, the continuous-time lower-bound rate is $r_{\text{cont}} = 2\alpha\lambda_{\min}$. As $\Delta t \rightarrow 0$ the Riemannian rate $r_{\text{Riem}} \rightarrow r_{\text{cont}}$, whereas the KL rate tends to $r_{\text{cont}}/4$: at small Δt the denominator of (15) tends to 2, so the decrement is $\frac{1}{2}\alpha\lambda_{\min}\Delta t$ and $r_{\text{KL}} \rightarrow \alpha\lambda_{\min}/2 = r_{\text{cont}}/4$. The KL contraction factor is thus a factor ≈ 4 more conservative than the Riemannian one even in the small-stepsize limit—this is the contraction-rate gap that survives the removal of the stepsize penalty.

Observed contraction. For each run we record the full energy-gap trajectory and report a terminal contraction factor $\hat{q}_{\text{term}} = (\text{gap}_N/\text{gap}_0)^{1/N}$ and rate $\hat{r}_{\text{term}} = -\log(\text{gap}_N/\text{gap}_0)/(N\Delta t)$, using a small floor (10^{-16}) on the raw gap for the logarithm (the raw gap is kept in the CSV, and floor-limited runs are flagged). We also fit a log-linear rate $\hat{r}_{\text{fit}}$ over the mid-regime $10^{-10} \leq \text{gap}_n/\text{gap}_0 \leq 10^{-2}$, excluding the machine-precision plateau. The *terminal slack* $q_{\text{theory}}^N \text{gap}_0/\text{gap}_N$ measures how far the method-specific theory envelope sits above the observed terminal gap.

Findings. Table 4 aggregates the benchmark and Figures 14–18 display it. Three observations hold consistently across both targets and all three λ . First, the observed contraction is *faster than either formula predicts*: in the rate scatter (Fig. 16) every point lies above the $y = x$ diagonal, and the terminal slack is positive by several orders of magnitude for every cell (Fig. 17, Table 4). Second, the KL contraction factor is the *more conservative* of the two: at every Δt on the grid $r_{\text{KL}}(\Delta t) \leq r_{\text{Riem}}(\Delta t)$, so the KL rate envelope predicts slower contraction and sits higher above the observed curve, and the observed/theory rate ratio is correspondingly larger for KL (Table 4). The terminal slack is large and positive for *both* schemes (several orders of magnitude); it is largest for KL at the finer fractions, while at the coarsest fractions the Riemannian observed gap descends to near machine precision (deeper observed convergence), which inflates the Riemannian slack into the KL range there. Third, at the finest fractions ($c = 0.05$) the two *observed* rates are nearly identical and close to the common continuous-time value, consistent with both schemes being first-order discretizations of the same flow; the Riemannian theoretical rate also approaches r_{cont} there (from below), while the KL theoretical rate approaches only $r_{\text{cont}}/4$ (eq. (15)), so both theoretical rates sit well below the observed rate and the KL one further still. The iteration-to-tolerance comparison (Fig. 18) tells the same story from the cost side: where both the observed and the theoretical iteration counts to gap $\leq 10^{-6}$ are finite, N_{theory} exceeds N_{obs} , and the KL theoretical count is the largest. The reading is now precise: both factors are genuine in-theorem contraction rates on this shared stepsize range, and the only theoretical distinction between the schemes is that the KL contraction factor is the more conservative of the two—a contraction-rate gap, not a stepsize gap.

8 Conclusion

The evidence supports a balanced reading, not a simple ranking.

- **The improved KL proof removes the artificial cubic stepsize penalty.** The KL relative-smoothness constant is now $\beta\lambda_{\text{max}}$, equal to the Riemannian one, so the two schemes share the theorem-safe stepsize $\Delta t \leq 1/(\beta\lambda_{\text{max}})$ (3). The KL scheme has no inherent stepsize disadvantage from the theory.
- **The Riemannian discretization is the geometrically faithful one.** Its update (5) is exactly a geodesic step of the affine-invariant geometry, with the cleanest interpretation as geodesic gradient descent. Its cost is the matrix exponential, immaterial here but the more expensive update in general.
- **Both schemes are stable well beyond the (shared) sufficient bound.** On the globally smooth targets each scheme is stable and monotone for stepsizes a couple of orders of magnitude larger than the shared sufficient condition $1/(\beta\lambda_{\text{max}})$ permits, with comparable empirical/theory ratios (Table 2, Fig. 7). The shared bound is conservative on these deterministic targets; this persists under the strict accuracy criterion, not just bare stability.
- **The KL covariance update is not the instability source.** For a log-concave target the KL precision $C^{-1} - \Delta t H$ is SPD for every $\Delta t > 0$, so the covariance step never leaves the SPD cone; the scalar diagnostic (Fig. 8) confirms unconditional contraction of the covariance to its fixed point.
- **Large-stepsize failures come from the shared mean update.** Both schemes diverge at large Δt through the explicit mean step $m_{n+1} = (1 - \Delta t C_n)m_n$, which overshoots once $\Delta t C_n > 2$ regardless of how the covariance is advanced. This is why the two schemes’ SPD

Table 4: Theorem contraction-rate benchmark on the shared theorem-safe grid, aggregated over $\lambda \in \{0.01, 0.1, 1\}$ (geometric means). For each target/method/ c : the observed terminal rate $\bar{r}_{\text{obs}}$, the method-specific theoretical rate $\bar{r}_{\text{th}}$, their ratio (observed faster when > 1), the terminal slack, the run count n , and the number of floor-limited runs. The observed/theory ratio is well above 1 throughout and is larger for KL than for Riemannian at matched c (the KL contraction factor being the more conservative); the terminal slack is large for both schemes.

target	method	c	$\bar{r}_{\text{obs}}$	$\bar{r}_{\text{th}}$	$\overline{r_{\text{obs}}/r_{\text{th}}}$	$\overline{\text{slack}}$	n	floor
Gaussian	riemannian	0.05	1.87	0.0979	19.1	6.5×10^9	3	0
		0.1	1.88	0.0958	19.6	8.2×10^9	3	0
		0.25	1.91	0.0892	21.4	1.7×10^{10}	3	0
		0.5	1.96	0.0776	25.3	7.3×10^{10}	3	0
		0.75	2.02	0.0653	31	4.1×10^{11}	3	0
		1	2.1	0.0524	40.1	3.4×10^{12}	3	0
	kl	0.05	1.87	0.0236	79.2	4.2×10^{11}	3	0
		0.1	1.88	0.0223	84.1	5.2×10^{11}	3	0
		0.25	1.9	0.0191	99.4	9.6×10^{11}	3	0
		0.5	1.93	0.0153	126	2.0×10^{12}	3	0
		0.75	1.92	0.0126	152	1.9×10^{12}	3	0
		1	1.9	0.0106	179	1.6×10^{12}	3	0
smooth	riemannian	0.05	1.63	0.0822	19.8	1.2×10^{10}	3	0
		0.1	1.63	0.0801	20.4	1.3×10^{10}	3	0
		0.25	1.63	0.0738	22.2	1.5×10^{10}	3	0
		0.5	1.64	0.0633	25.9	1.9×10^{10}	3	0
		0.75	1.65	0.0527	31.2	2.5×10^{10}	3	0
		1	1.65	0.0422	39.2	3.2×10^{10}	3	0
	kl	0.05	1.63	0.02	81.3	3.0×10^{10}	3	0
		0.1	1.63	0.0191	85.2	3.0×10^{10}	3	0
		0.25	1.62	0.0167	97.1	2.9×10^{10}	3	0
		0.5	1.61	0.0138	117	2.7×10^{10}	3	0
		0.75	1.6	0.0117	137	2.5×10^{10}	3	0
		1	1.6	0.0101	158	2.3×10^{10}	3	0

and stability boundaries coincide on the quartic target and why the absolute stepsize ceiling is set by the mean, not the covariance.

- **At a matched stepsize the Riemannian scheme usually converges faster.** On both globally smooth targets the geodesic update reaches a smaller terminal energy gap than KL at every convergent stepsize, with the margin growing from a small factor at fine Δt to 10^4 – 10^8 at the coarsest convergent stepsizes (Sec. 6, Table 3); KL leads only on the non-smooth quartic and only modestly. So when one operates at a stepsize chosen as large as stability allows—the practical regime—the Riemannian update is typically the faster choice.

We are careful not to overclaim. The two schemes now share one theorem-safe stepsize, and their absolute stable stepsizes are comparable (indeed at $\lambda = 1$ the Riemannian scheme tolerates a slightly larger stable Δt than KL), while on the non-smooth quartic target both are restricted to small stepsizes, where KL is the faster of the two. The conclusions are specifically that the improved

KL proof leaves *no theoretical stepsize disadvantage* for the KL scheme, that the only remaining theoretical distinction is the *contraction factor* (more conservative for KL), and that there is a *matched-stepsize convergence-speed advantage of the Riemannian update on smooth targets*—not a blanket statement that one scheme dominates the other in every respect.

Limitations. This is numerical evidence on $d \leq 2$ deterministic targets, not a theorem. The stepsize grid is coarse (factors of 2–2.5), so the reported empirical maxima are accurate only to within one grid step; the qualitative gap between empirical stability and the shared sufficient bound is larger than this resolution. The improved KL proof already establishes that the KL scheme admits the same theorem-safe stepsize as the Riemannian one; the present data show that this shared bound is conservative in practice and that the shared mean step, not the covariance update, is the true stepsize bottleneck.

A Figures

All figures are collected here to keep the main text readable; they are referenced by number from the sections above and appear in the order first cited.

A.1 Figures for the stability results (Sec. 5)

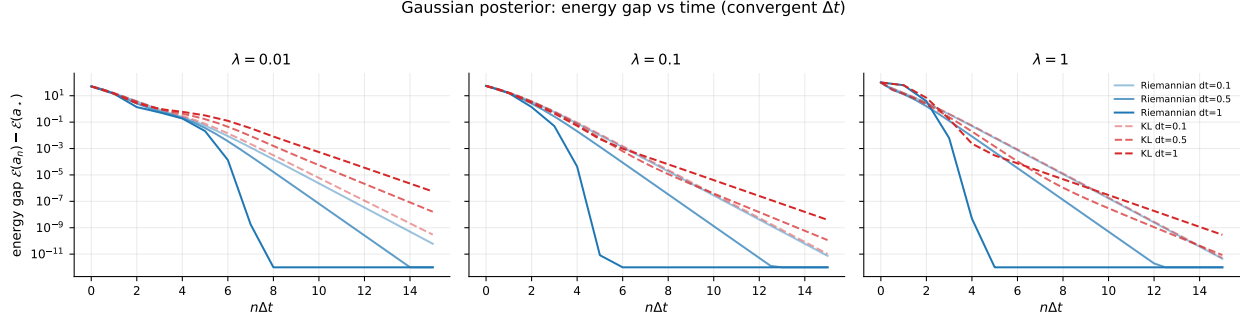


Figure 1: Gaussian posterior: energy gap vs elapsed time $n\Delta t$, $\lambda \in \{0.01, 0.1, 1\}$, Riemannian (blue) vs KL (red) at $\Delta t \in \{0.1, 0.5, 1\}$ (darker = larger Δt). The analytic gap (9) reaches machine precision.

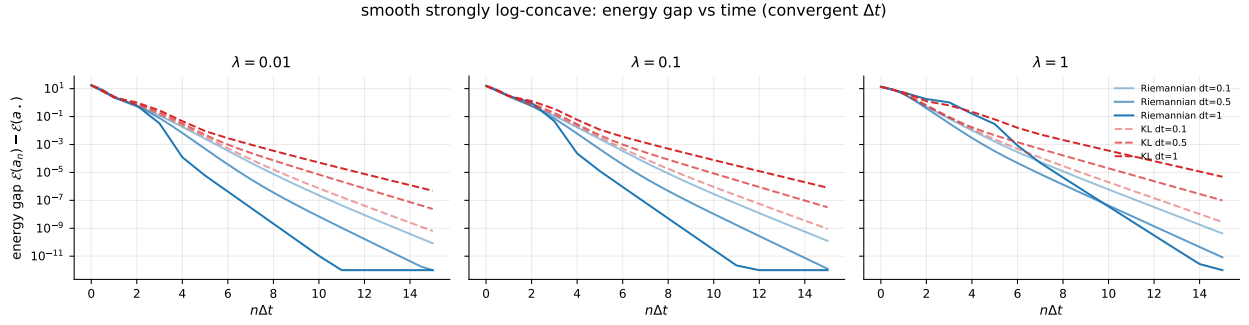


Figure 2: Smooth strongly log-concave posterior: energy gap vs elapsed time $n\Delta t$. Both schemes converge at the representative stepsizes; the gap is measured against the numerically optimized a_* .

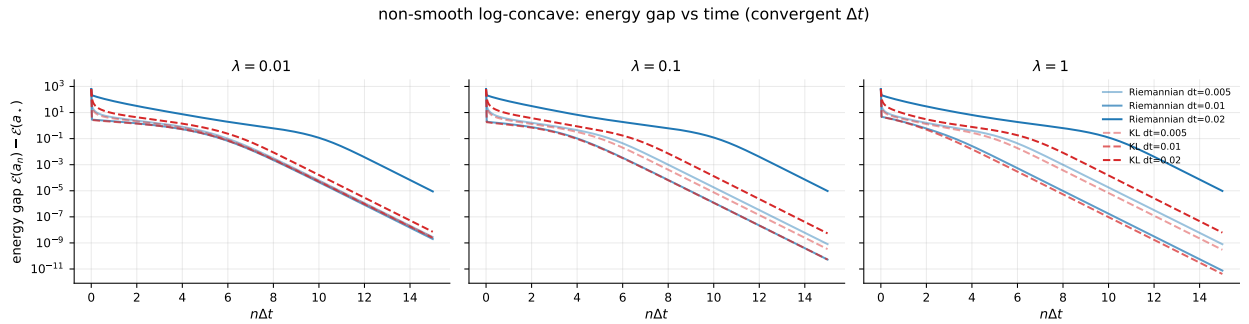


Figure 3: Non-smooth (quartic) log-concave posterior: energy gap vs elapsed time $n\Delta t$ at its convergent representative stepsizes $\Delta t \in \{0.005, 0.01, 0.02\}$. Both schemes converge cleanly here; the stepsizes are an order of magnitude smaller than for the smooth targets, reflecting the unbounded curvature of the quartic tail. (Larger stepsizes diverge for both methods and are excluded; see the stability heatmap, Fig. 6.)

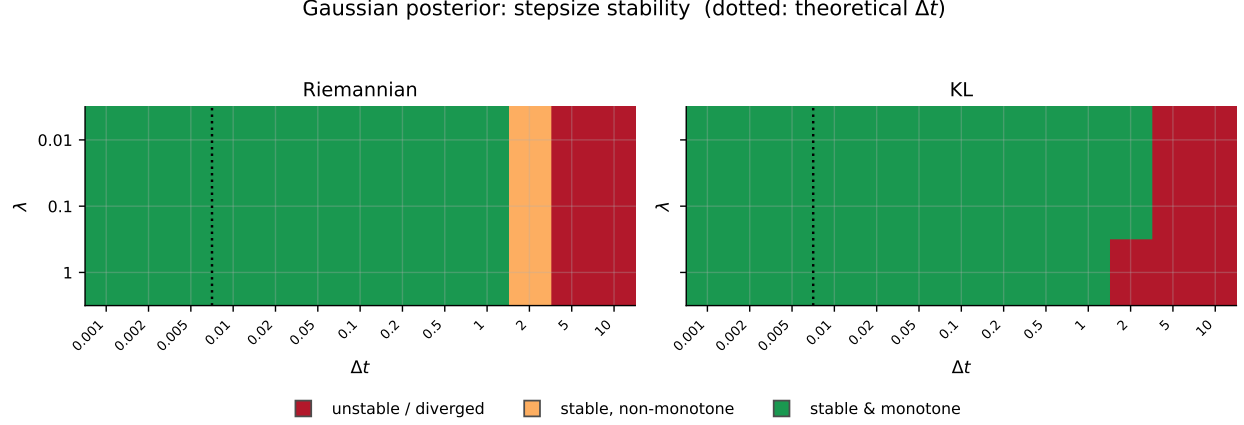


Figure 4: Gaussian posterior stability classification. Green: stable & monotone; amber: stable, non-monotone; red: unstable/diverged. Dotted line: the shared theoretical $\Delta t = 1/(\beta\lambda_{\max})$ (identical for both methods). Both stable regions extend well past it.

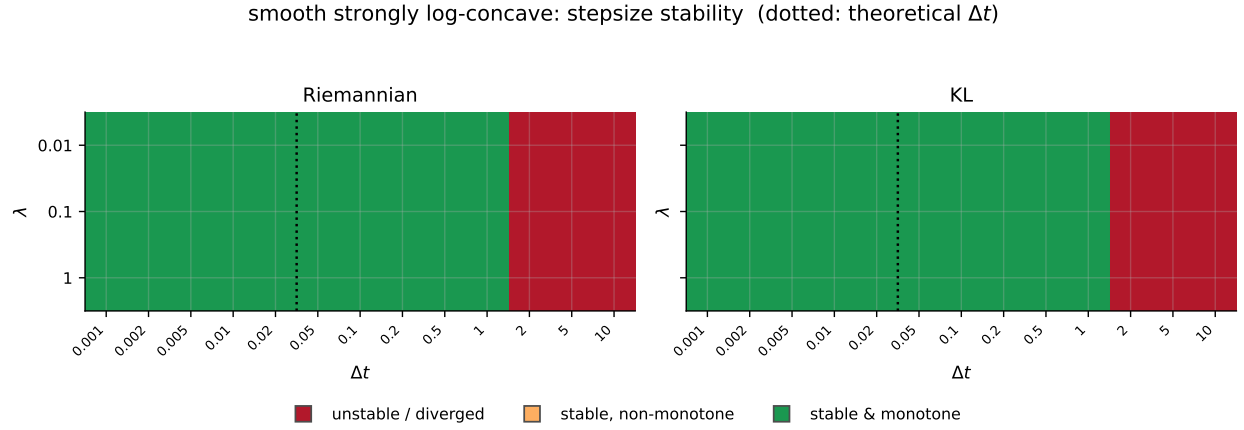


Figure 5: Smooth log-concave stability classification (same legend). The shared theoretical line ($\sim 10^{-1}$) sits well to the left of the empirical boundary (~ 1) for both methods.

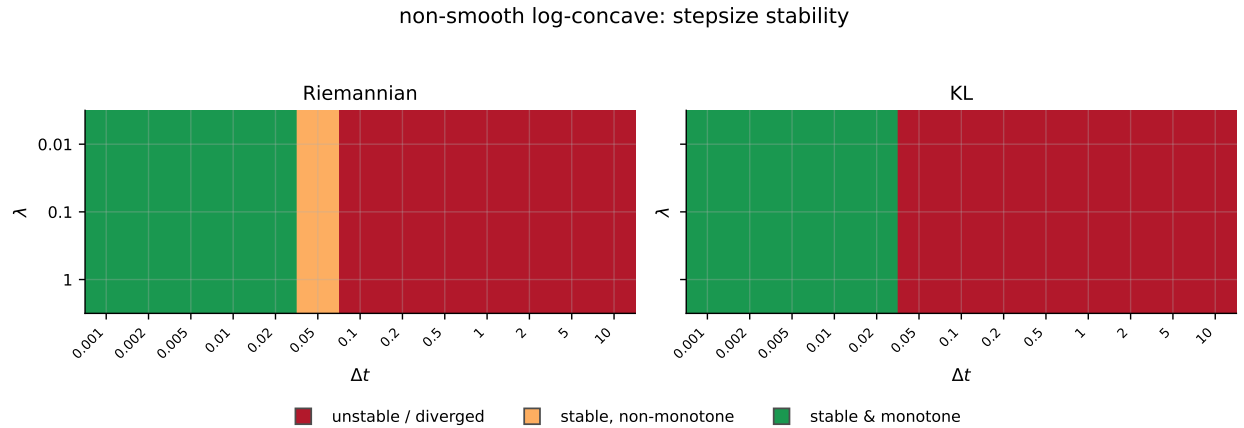


Figure 6: Non-smooth log-concave stability classification (no theory line, since the target is not globally smooth). Both schemes are stable only for small Δt .

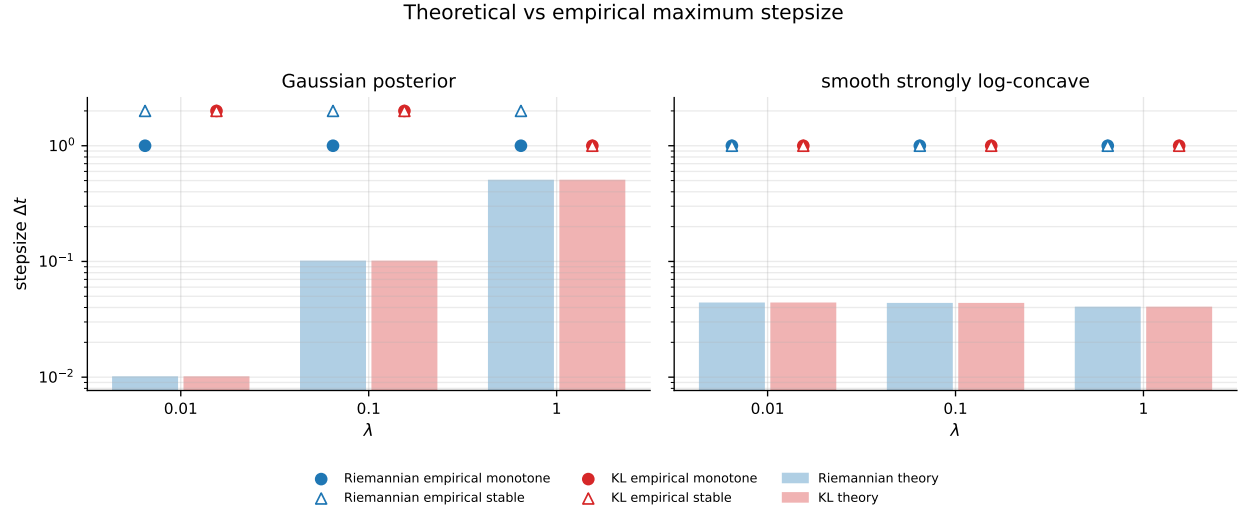


Figure 7: Theoretical (shaded bars) vs empirical maximum stepsize (markers: $\circ$ monotone, Δ stable) for the Gaussian and smooth targets, Riemannian (blue) vs KL (red), log scale. The theoretical bar is the *same* $1/(\beta\lambda_{\max})$ for both methods, and each scheme's empirical markers sit a comparable, modest factor above it.

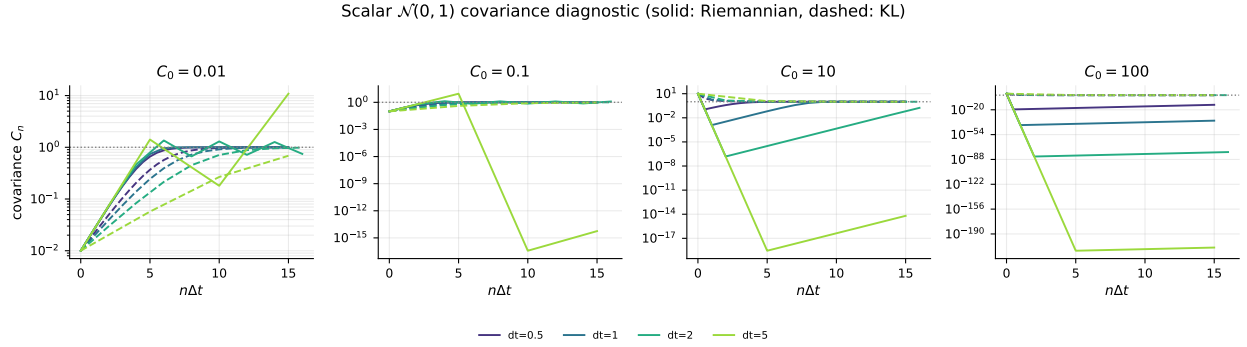


Figure 8: Scalar $\mathcal{N}(0, 1)$ covariance diagnostic. Solid: Riemannian; dashed: KL; color shade encodes Δt ; dotted grey: target $C = 1$. The KL covariance contracts to 1 for all $(C_0, \Delta t)$ shown.

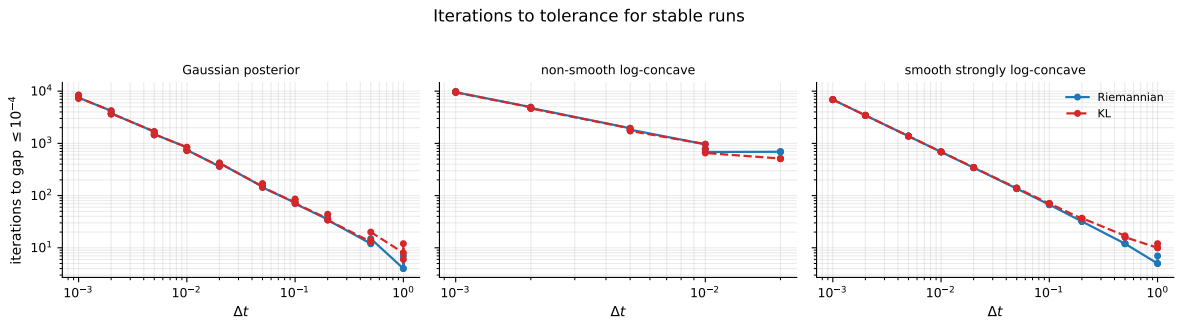


Figure 9: Iterations to reach gap $\leq 10^{-4}$ for stable runs, vs Δt (log-log). At matched Δt the two schemes are comparable; the meaningful difference is the range of stable Δt , not per-step cost.

A.2 Figures for the convergence-speed study (Sec. 6)

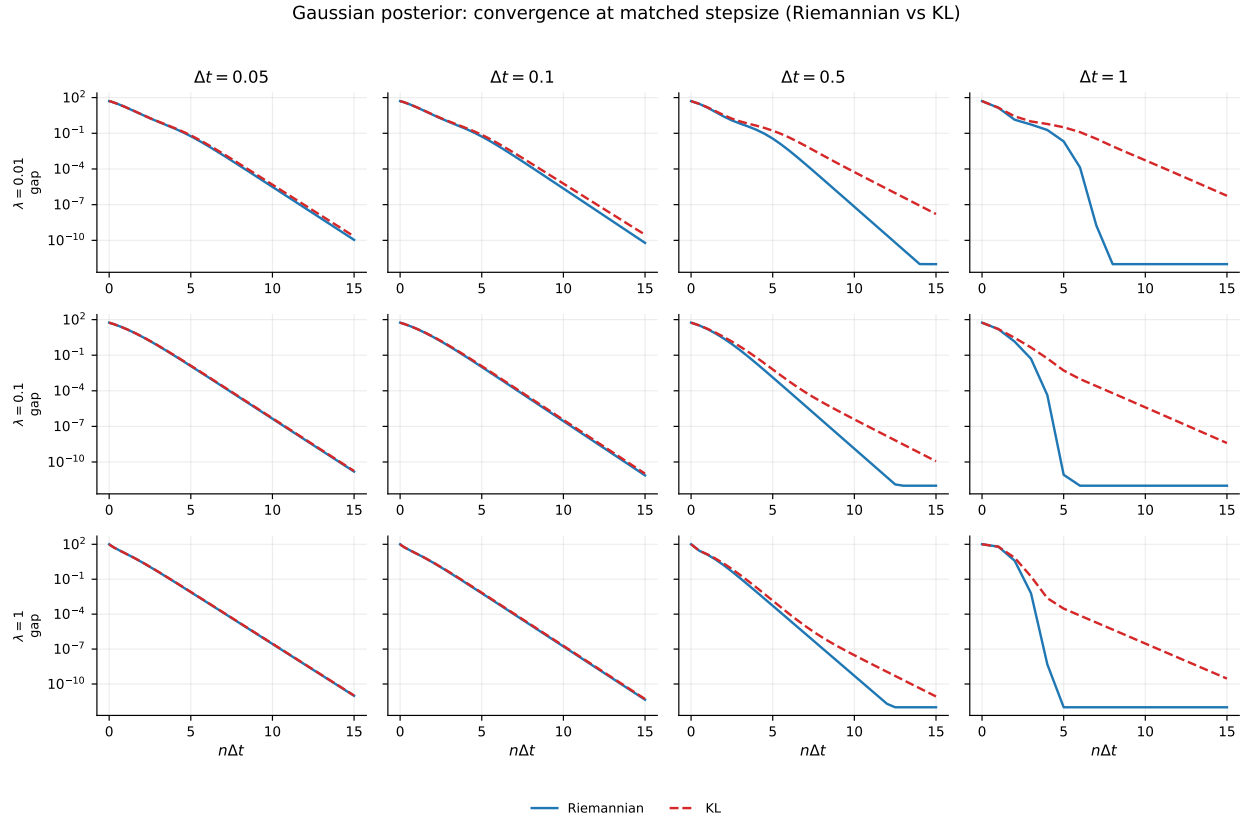


Figure 10: Gaussian posterior: energy gap vs elapsed time $n\Delta t$, Riemannian (solid) vs KL (dashed). Rows: λ ; columns: convergent stepsize Δt . The two schemes coincide at the fine stepsizes (left) but separate at the coarse stepsizes (right), where the Riemannian update reaches machine precision while KL plateaus higher.

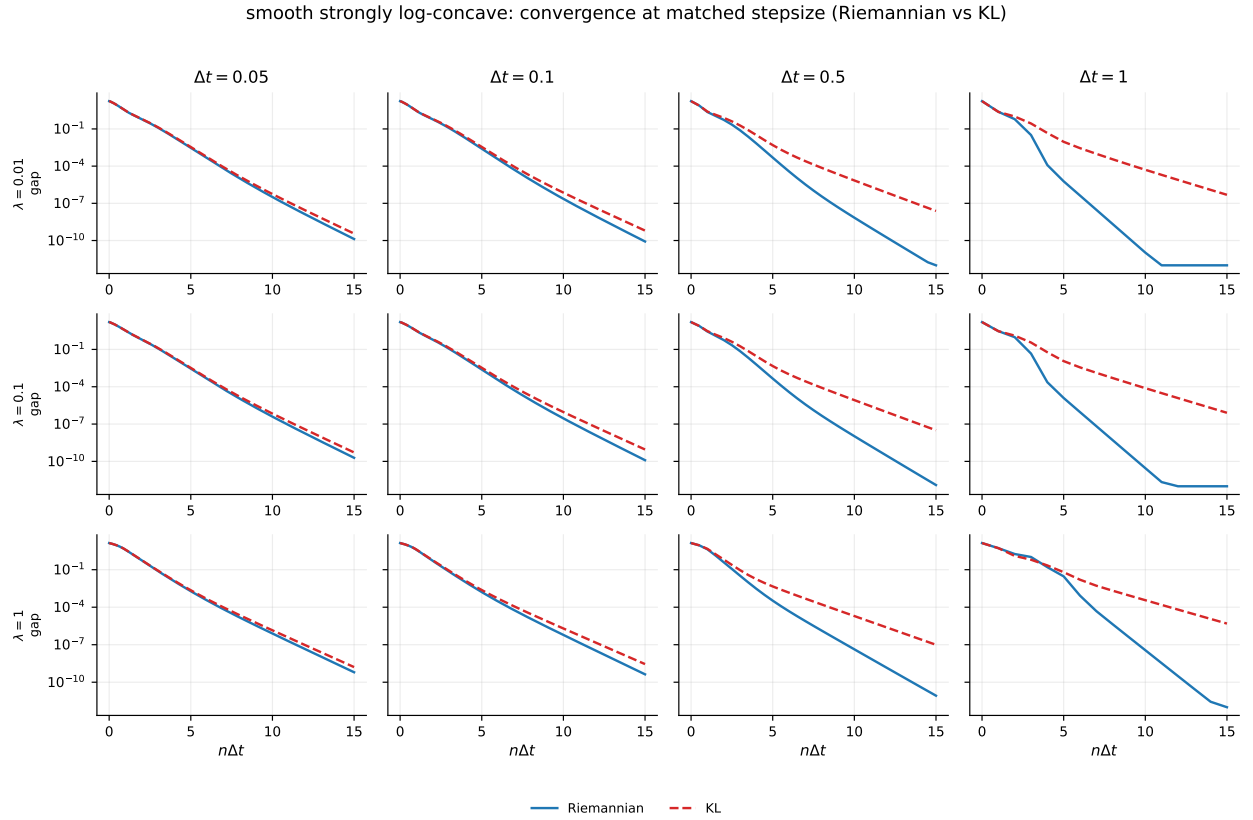


Figure 11: Smooth strongly log-concave posterior: energy gap vs elapsed time $n\Delta t$ (same layout and legend as Fig. 10). The same regime split holds: matched-stepsize agreement at fine Δt , and a growing Riemannian advantage as Δt coarsens.

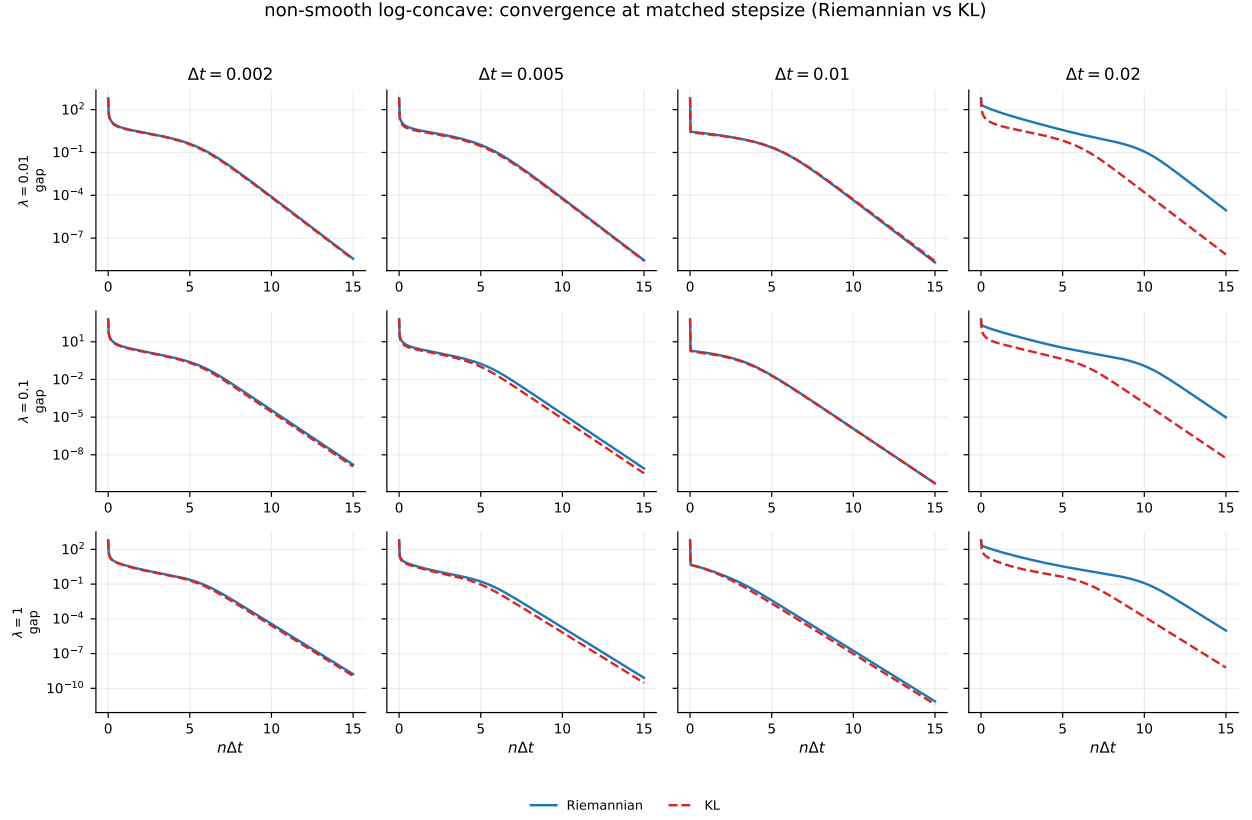


Figure 12: Non-smooth (quartic) log-concave posterior: energy gap vs elapsed time $n\Delta t$ (same layout). The convergent stepsizes are an order of magnitude smaller than for the smooth targets; the schemes agree closely at the finer stepsizes, but here *KL* holds the advantage—most clearly at the coarsest convergent $\Delta t = 0.02$, where the Riemannian exponential map sits near its stability edge (terminal gap $\sim 10^{-5}$) while *KL* has converged ($\sim 10^{-9}$). This is the one target on which *KL*, not Riemannian, is the faster scheme.

Cost of a stepsize: iterations $\propto 1/\Delta t$, elapsed $n\Delta t$ roughly flat

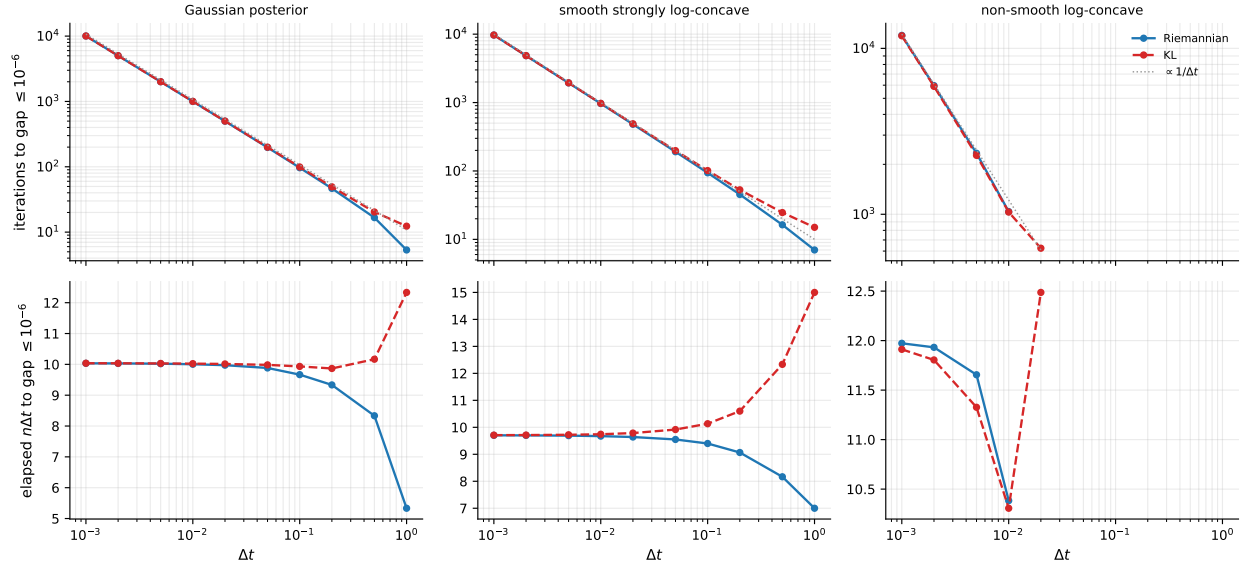


Figure 13: Cost to reach gap $\leq 10^{-6}$ for stable runs, averaged over λ , vs Δt . Top: iterations (log-log), with the $\propto 1/\Delta t$ reference slope; bottom: elapsed time $n\Delta t$ (semi-log x), nearly flat in Δt . Riemannian (blue) and KL (red) coincide at fine Δt and separate only at the coarsest convergent stepsizes; the faster scheme is target-dependent (Riemannian on the smooth targets, KL on the non-smooth one).

A.3 Figures for the theoretical-rate benchmark (Sec. 7)

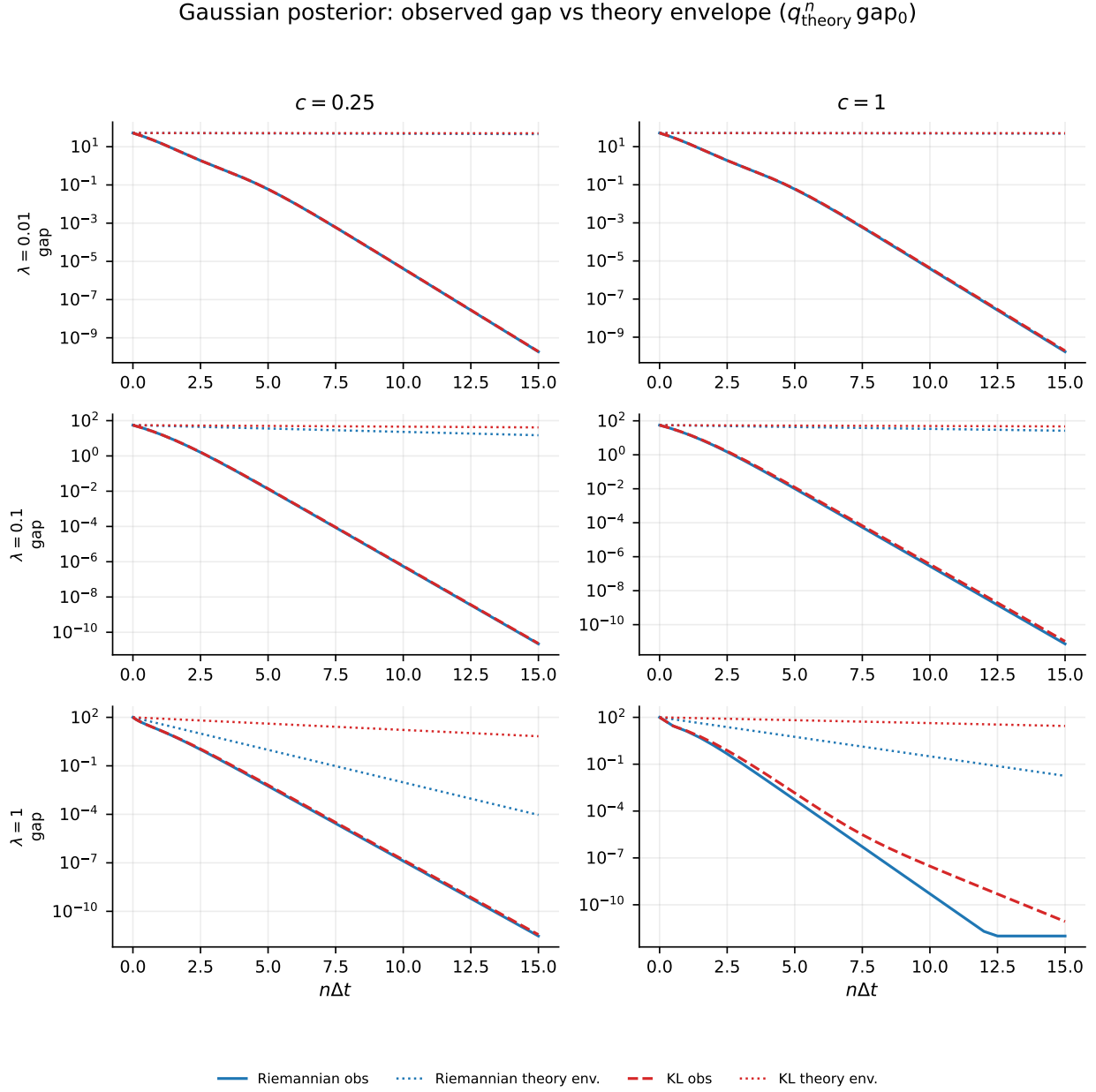


Figure 14: Gaussian posterior: observed energy gap (solid Riemannian, dashed KL) versus the method-specific theory envelopes $q_{\text{theory}}^n \text{gap}_0$ (dotted), rows λ , columns $c = \Delta t / \Delta t_{\text{ref}}$. Both envelopes sit well above the observed curves; the KL envelope (dotted red) is the higher of the two.

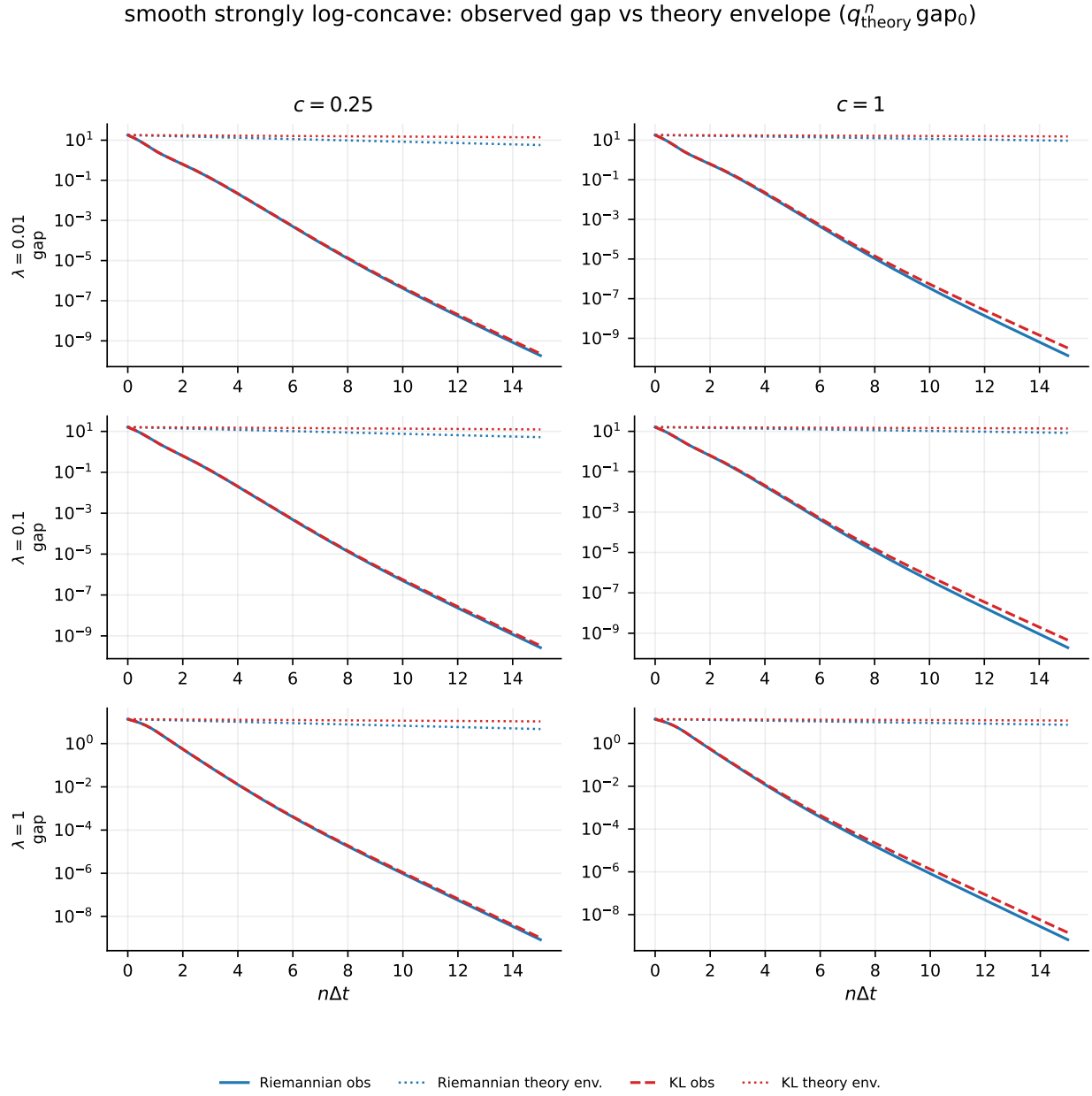


Figure 15: Smooth strongly log-concave posterior: observed gap vs theory envelopes (same layout and legend as Fig. 14). The same ordering holds: observed contraction is faster than both envelopes, with the KL envelope the more conservative.

Observed vs theoretical contraction rate (above $y = x$: faster than theory)

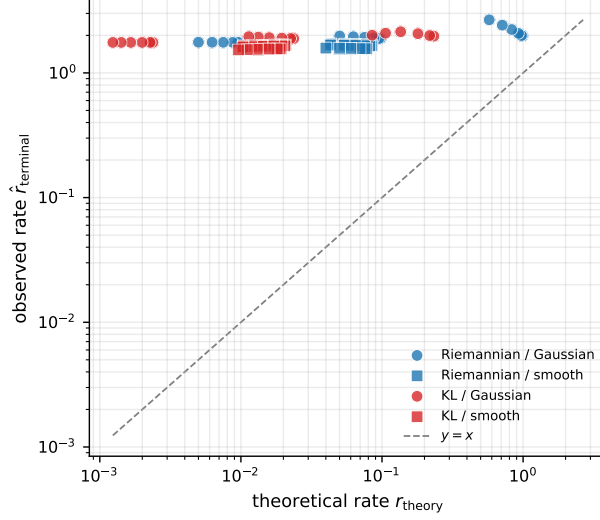


Figure 16: Observed terminal rate $\hat{r}_{\text{term}}$ vs method-specific theoretical rate r_{theory} (log-log), all target/ λ/c runs. Color: method; marker: target. Every point lies above $y = x$: the observed contraction is faster than the theorem prediction, and the KL points (red) sit furthest to the left (smallest theoretical rate) at matched observed rate.

Terminal slack (theory bound / observed gap); larger = more conservative
(* = observed gap at numerical floor)

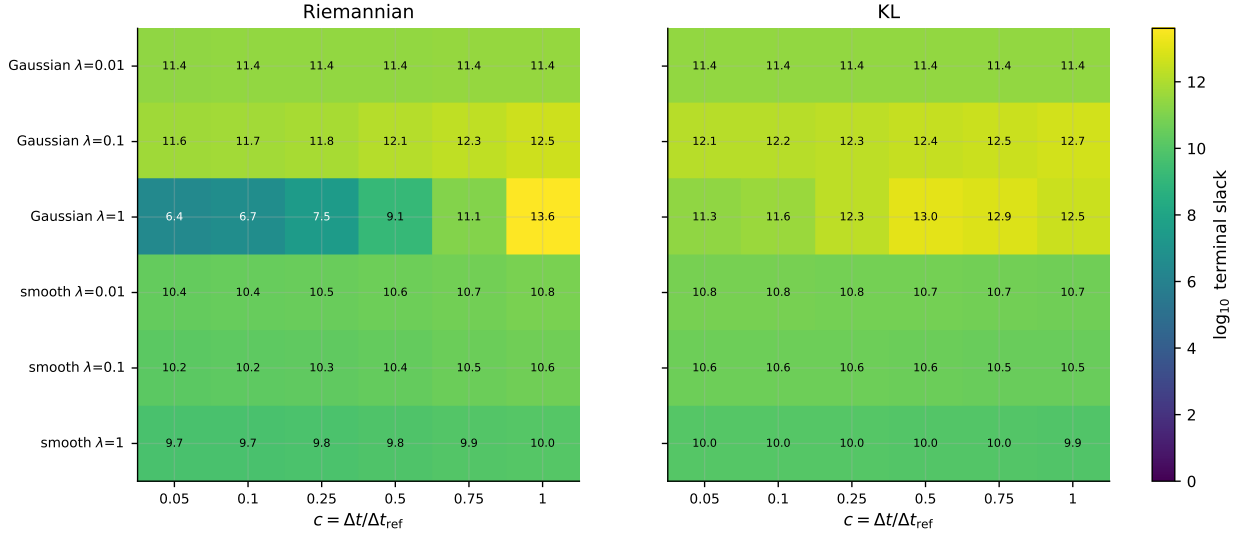


Figure 17: $\log_{10}$ terminal slack $q_{\text{theory}}^N \text{gap}_0 / \text{gap}_N$ per target/ λ/c , Riemannian (left) vs KL (right). Larger is more conservative. The slack is large and positive throughout for both schemes; it is largest for KL at the finer fractions, while at the coarsest fractions the Riemannian observed gap reaches near machine precision (deeper observed convergence) and its slack rises into the KL range. A trailing * marks a run whose observed gap reached the numerical floor (slack is then a lower bound).

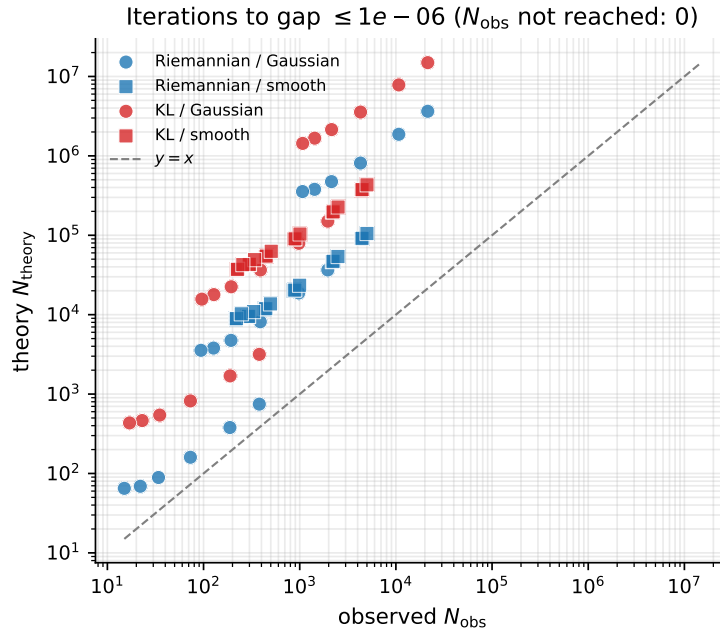


Figure 18: Theoretical vs observed iterations to gap $\leq 10^{-6}$ on the common grid. Points plotted only where both are finite; the count of runs whose *observed* trajectory does not reach 10^{-6} within T is annotated. Where both are defined, $N_{\text{theory}} > N_{\text{obs}}$ (above $y = x$), most strongly for KL.